7-2 Computer Software

Point!

Hardware and Software

(1) Hardware : Devices such as the main computer unit and peripheral devices.
<Examples> CPU, memory, storage, etc.

(2) ([1]**Software**) : Programs and data that operate on hardware.

[1] ([2]**System software**) : Software necessary for operating the hardware.

[2] ([3]**Operating system (OS)**) : A type of system software responsible for the basic functions of a computer. An OS has management functions such as task management, memory management, and file management.
<Examples> Windows, macOS, Android OS, etc.

[3] ([4]**Application software**) : Software that operates on top of system software. Also known as application software.
<Examples> Word processing software, spreadsheet software, etc.

(3) ([5]**Device driver**) : A program that controls communication between a connected device and software on a computer. Also known simply as a "driver."

Warm Up

For the following statements **A** to **D** about an OS, mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A Software used to perform specific tasks.
B Smartphones do not come equipped with an OS.
C A type of system software essential for operating the hardware.
D Application software cannot run without system software.

Explanation

A: An explanation about application software. Therefore, ×
B: Smartphones are equipped with an OS. For example, iPhones are equipped with iOS, while Android phones run on Android OS. Therefore, ×
C: An OS is a type of system software responsible for the basic functions of a computer. Therefore, ○
D: Application software is software that operates on top of system software. Therefore, ○

Try

Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [3] with the appropriate terms.

> Types of software include ([1]) software and ([2]) software. ([1]) software includes ([3]). In the case of computers, examples are Windows and macOS. Additionally, ([2]) software includes word processing software and spreadsheet software.

(2) What is the term for software that is used to control and operate devices connected to a computer?

(3) For the following statements **A** to **D**, answer with "a" if the statement is about operating systems or "b" if the statement is about application software.

A A program executed on a computer that is designed for a specific purpose.
B A program that controls software and hardware on a computer system.
C Includes word processing software and spreadsheet software.
D Included in the system software.

Exercise

1 Answer the following questions.

(1) Choose all the devices and software corresponding to the following [1] to [3] from the options **A** to **F** below, and answer using the letters.

[1] Operating system [2] Hardware [3] Application software

A Memory **B** Spreadsheet software **C** Windows
D macOS **E** CPU **F** Document processing software

(2) Choose one correct statement regarding device drivers from the options **A** to **D** below, and answer using the letter.

A A type of system software. Examples include, Windows, macOS, etc.
B Also referred to as application software.
C A program that controls communication between the devices connected to a computer and the software.
D The software necessary for operating hardware.